

Numerical Analysis homework 1

Zirui Chen 22435036 *

(24), Zhejiang University

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I

I-a

prove:

we know the lagrange interpolation of f is $p_1(f; x) = f(x_0) \frac{x-x_1}{x_0-x_1} + f(x_1) \frac{x-x_0}{x_1-x_0} = \frac{3}{2} - \frac{x}{2}$

then we can use the number of $x_0, x_1, f(x) - p_1(f; x) = \frac{1}{2x}(x-1)(x-2) \Rightarrow f''(\xi(x)) = \frac{1}{x} \Rightarrow \xi(x) = \sqrt[3]{2x}$

□

I-b

Prove:

Follows from above, we have: $\xi(x) = \sqrt[3]{2x}, \max_{x \in [1,2]} \xi(x) = \sqrt[3]{4}, \min_{x \in [1,2]} \xi(x) = \sqrt[3]{2}, \max_{x \in [1,2]} f''(\xi(x)) = 1$

□

II

Prove:

Use lagrange interpolation $p_n f; x = \sum_{k=0}^n f_k(x) \left(\frac{\pi_{n+1}(x)}{(x-x_k)\pi'_{n+1}(x_k)} \right)^2$

it is obvious that $p_n(f; x) \in \mathbb{P}_{2n}^+$

□

III

III-a

Prove:

To prove the induction, we begin with the base case $n=0, f[t] = f(t) = e^t$

Assume that the induction hypothesis is true for $n=k-1$, We need to show that the statement holds for $n=k$.

We have:

$$f[t, t+1, \dots, t+n] = \frac{1}{t+n-t} (f[t+n, \dots, t+1] - f[t+n-1, \dots, t]) \quad (1)$$

$$= \frac{1}{n} \left(\frac{(e-1)^{n-1}}{(n-1)!} e^{t+1} - \frac{(e-1)^{n-1}}{(n-1)!} e^t \right) \quad (2)$$

$$= \frac{(e-1)^{n-1}}{n!} e^t \quad (3)$$

□

*Electronic address: 18107732576@163.com

III-b

Prove:

we know $f^{(n)}(x) = e^x$ and $f[t, t+1, \dots, t+n] = \frac{(e-1)^n}{n!} e^t$.
 $\Rightarrow f[0, 1, \dots, n] = \frac{(e-1)^n}{n!} \Rightarrow \xi = n!n(e-1)$, because $e-1 > \sqrt[n]{e} \Rightarrow \xi > \frac{n}{2}$.

□

IV

IV-a

Newton fomular:

$$p_3(f; x) = f[0] + f[0, 1](x-0) + f[0, 1, 3] + x(x-1) + f[0, 1, 3, 4]x(x-1)(x-3) \quad (4)$$

$$= 5 - 2x + x(x-1) + \frac{1}{4}x(x-1)(x-3). \quad (5)$$

IV-b

Find minimum :

we just need to find the minimum of $p_3(f; x)$, then $x_{min} = \sqrt{5}$.

□

V

V-a

Compute $f[0, 1, 1, 1, 2, 2]$:

0		0					
1		1	1				
1		1	7	6			
1		1	7	21	15		
2		128	127	120	99	42	
2		128	448	321	201	102	30

So , $f[0, 1, 1, 1, 2, 2] = 30$.

V-b

Determine ξ :

Newton interpolation of x^7 : $p(f; x) = 0 + x + 6x(x-1) + 15x(x-1)(x-1)^2 + 42x(x-1)^3 + 30x(x-1)^3(x-2) + \frac{f^{(6)}(\xi)}{6!}$.

It follows from Corollary 2.22 that $f[x_0, x_1, \dots, x_n] = \frac{1}{n!} f^{(n)}(\xi) \Rightarrow \xi = 1.195$.

VI

VI-a

Estimate :

First we give the Hermite interpolation formula :

$$p_4(f; x) = 1 + x - 2x(x-1) + \frac{2}{3}x(x-1)^2 - \frac{5}{36}x(x-1)^2(x-3)$$

, Then we get $f(2) \approx p_4(f; 2) = \frac{11}{18}$.

VI-b

Estimate :

Follows from the Theorem 2.37 we have ,

$$f(x) - p_4(f; x) = \frac{f^5(\xi)}{5!}(x-1)(x-2)^2(x-3), \quad (6)$$

$$\Rightarrow f(2) - p_4(f; 2) = \frac{f^5(\xi)}{5!} \leq \frac{M}{60}. \quad (7)$$

VII

Prove :

Assume $x_0 = x, x_1 = x + h$,the induction basis holds for $n = 1$,because

$$\triangle f(X) = f(x+h) - f(x) = f[x_1] - f[x_0] = hf[x_0, x_1]$$

. Suppose the induction hypothesis holds for $n - 1$, we need to show that it holds for n .

$$\triangle^n f(x) = \triangle^{n-1} \triangle f(x) = \triangle^{n-1}(f(x+h) - f(x)) \quad (8)$$

$$= \triangle^{n-1} f(x+h) - \triangle^{n-1} f(x) = (n-1)! h^{n-1} (f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]) \quad (9)$$

$$= (n-1)! \cdot h^{n-1} (x_n - x_0) f[x_0, \dots, x_n] = n! \cdot h^n f[x_0, \dots, x_n]. \quad (10)$$

The similar result holds for $\nabla^n f(x) = n! \cdot h^n f[x_0, \dots, x_{n-k}]$.

□

VIII

Prove :

Notice that

$$\frac{\partial}{\partial x_0} f[x_0, \dots, x_n] = \lim_{\nabla x \rightarrow 0} \frac{1}{\nabla x} (f[x_0 + \nabla x, \dots, x_n] - f[x_0, \dots, x_n]) \quad (11)$$

$$= \lim_{\nabla x \rightarrow 0} f[x_0 + \nabla x, x_0, \dots, x_n] \quad (12)$$

$$= f[x_0, x_0, \dots, x_n]. \quad (13)$$

It is not sure about other variables ,because we dont know the differential of the variable.

□

IX

Determine :

First we denote $p(x)$ as the polynomial which the problem asked , consider $x = (\frac{b-a}{2})y + \frac{a+b}{2}$,which $y \in [-1, 1]$. Then the polynomial $p(y)$ has the leading coefficient $a_0(\frac{b-a}{2})^n$,a chebyshev polynomial of degree n in $[-1, 1]$ has maximum number 1. Then we follow from the Theorem 2.47 to get the minimum maximum value of $p(y)$ is $a_0 \frac{(b-a)^n}{2^{2n-1}}$. It is also the minimum maximum value of $p(x)$.

X

Prove :

Transform the chebyshev polynomial's domain from $[-1, 1]$ to $\mathbb{R}$,we have:

$$T_0(x) = 1, T_1(x) = x, T_2(x) = 2x^2 - 1, T_3(x) = 4x^3 - 3x, \dots, T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$$

, if $\exists p \in \mathbb{P}_n^a$,such that $\|p\|_\infty < \|\widehat{p}_n(x)\|_\infty$, Consider $Q(x) = \widehat{p}_n(x) - p(x)$,then we know that $Q(x)$ is a polynomial which has n zero points in $[-1, 1]$, and it is obvious that $Q(a) = 0$. But it constrat to $Q(x)$ is a n degree polynomial in $\mathbb{R}$.

□

XI

Prove :

Use the porperty of Berstain polynomials,

$$\frac{n-k}{n}b_{n,k}(t) + \frac{k+1}{n}b_{n,k+1}(t) = \frac{n-k}{n} \binom{n}{k} t^k (1-t)^{n-k} + \frac{k+1}{n} \binom{n}{k+1} t^{k+1} (1-t)^{n-k-1} \quad (14)$$

$$= t^k (1-t)^{n-k-1} \left(\frac{n-k}{n} \frac{n!}{k!(n-k)!} (1-t) + \frac{k+1}{n} \frac{n!}{(k+1)!(n-k-1)!} t \right) \quad (15)$$

$$= t^k (1-t)^{n-k-1} \binom{n-1}{k} \quad (16)$$

$$= b_{n-1,k}(t). \quad (17)$$

□

XII

Prove :

For $\forall k = 0, 1, 2, \dots, n$, we have $\int_0^1 b_{n,k}(t) dt = \frac{1}{n+1}$.

$$\int_0^1 \frac{n!}{k!(n-k)!} t^k (1-t)^{n-k} dt = \frac{n!}{k!(n-k)!} \cdot \frac{n-k}{k+1} \int_0^1 t^{k+1} (1-t)^{n-k-1} dt \quad (18)$$

$$= \frac{n!}{k!(n-k)!} \cdot \frac{(n-k)(n-k-1)}{(k+1)(k+2)} \int_0^1 t^{k+2} (1-t)^{n-k-2} dt \quad (19)$$

$$= \dots = \frac{n!}{k!(n-k)!} \cdot \frac{(n-k)!k!}{n!} \int_0^1 t^n dt \quad (20)$$

$$= \frac{1}{n+1} \quad (21)$$

□

Acknowledgement

Give your acknowledgements here(if any).

References

- [1] Qinghai Zhang and Zhixuan Li. “Boolean algebra of two-dimensional continua with arbitrarily complex topology”. In: *Math. Comput.* 89 (2020), pp. 2333–2364. URL: <https://api.semanticscholar.org/CorpusID:214531793>.

If you are not familiar with **biblatex**, [1] it is acceptable to put a table here for your references.